

A decorative pattern of interconnected hexagons in dark blue and light blue, located on the left side of the slide.A decorative pattern of circuit-like lines and dots in dark blue, located on the right side of the slide.

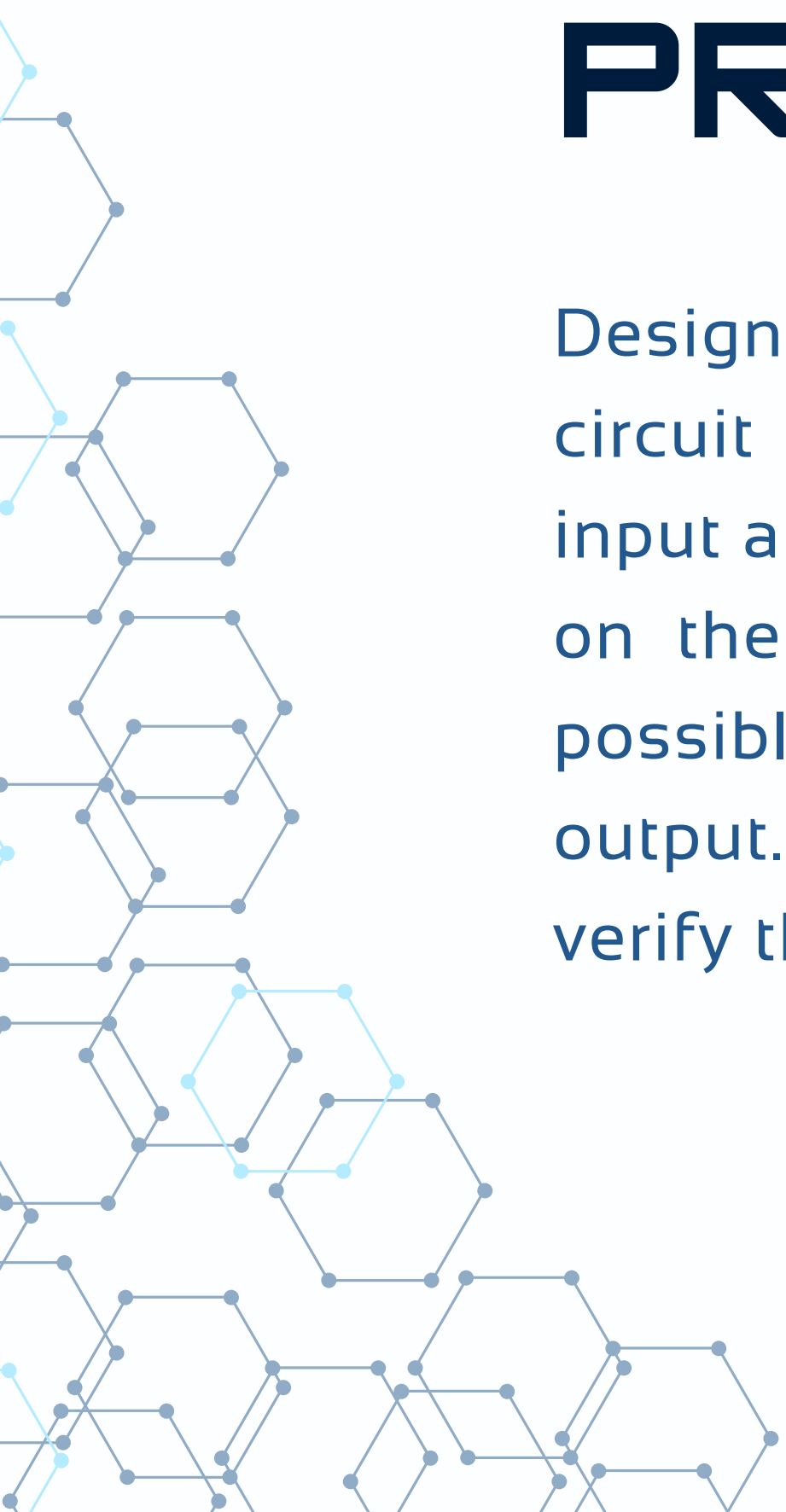
1:8 DEMULTIPLEXER

DESIGN AND SIMULATION USING VERILOG HDL

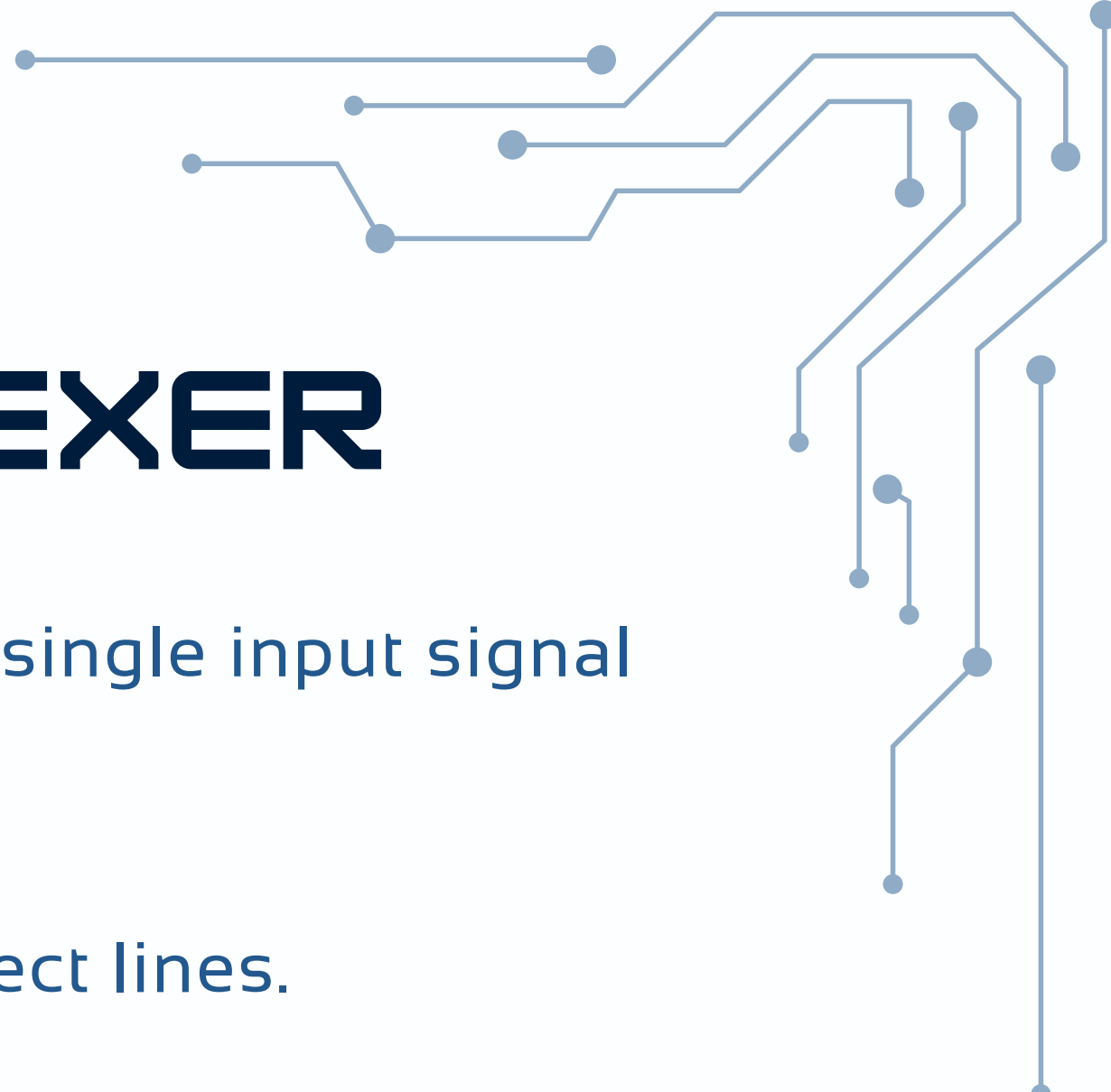
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PROBLEM STATEMENT



Design and simulate a 1:8 Demultiplexer using Verilog HDL. The circuit should accept a single 1-bit data input and a 3-bit select input and route the input signal to one of eight output lines based on the select input. Create a testbench that demonstrates all possible select combinations and verifies the corresponding output. Explain the working principle of the demultiplexer and verify the functionality using simulation in Vivado.



WHAT IS A DEMULTIPLEXER

A demultiplexer, or DEMUX, is a digital circuit that takes a single input signal and routes it to one of many output lines.

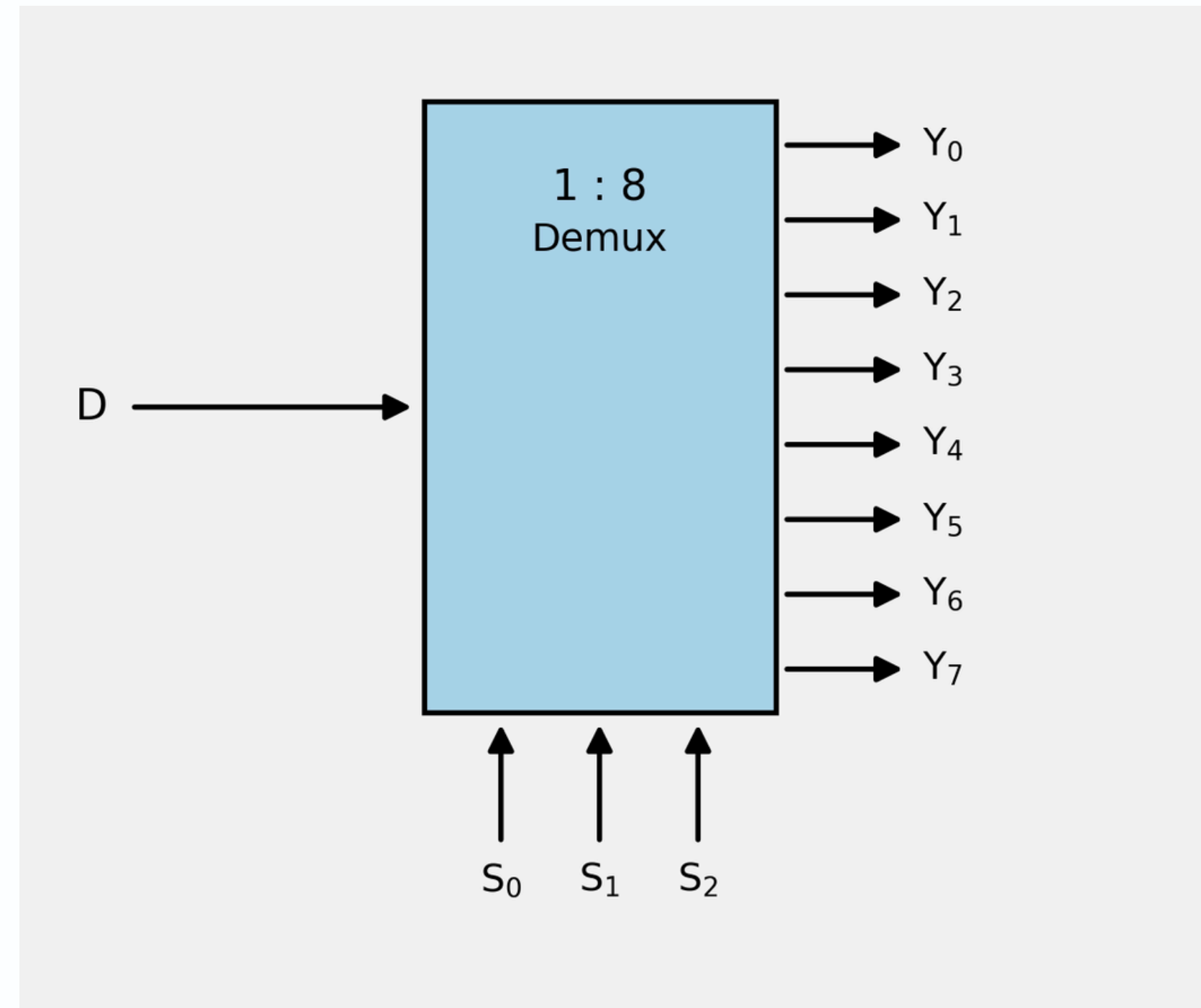
1:8 DEMUX

A 1:8 Demux routes 1 input to 1 of 8 outputs based on 3 select lines.

- 1 data input: **D**
 - 3 select inputs: **S_2, S_1, S_0**
 - 8 outputs: **$Y_0 - Y_7$**
 - As **$2^3 = 8$** , the 3 select lines can select 1 of 8 outputs.
 - The selected output carries the input signal.
 - All other outputs remain 0.
- 

WORKING PRINCIPLE OF A DEMUX

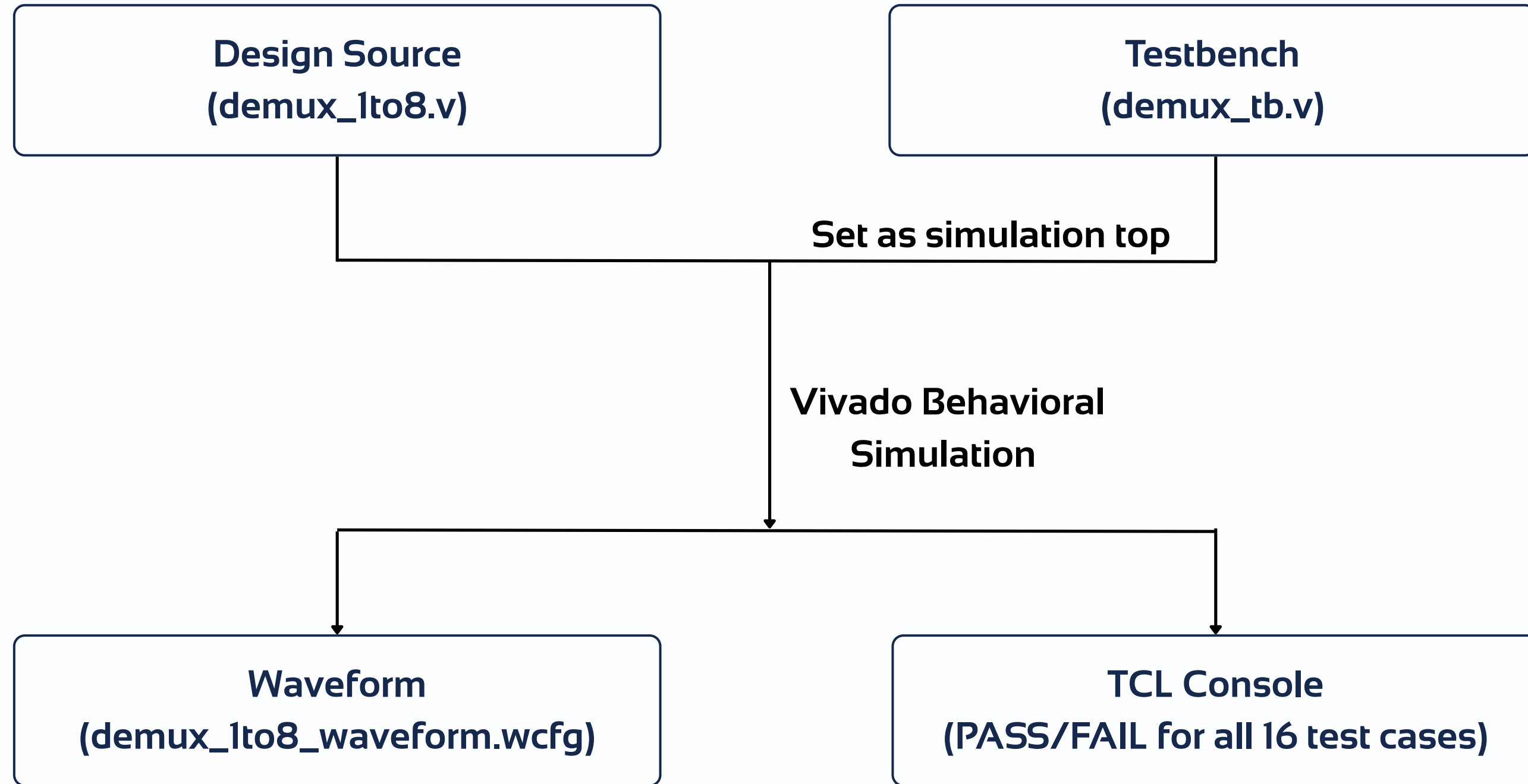
SELECT ($S_2S_1S_0$)	OUTPUT CARRYING D
000	$Y_0 = D$
001	$Y_1 = D$
010	$Y_2 = D$
011	$Y_3 = D$
100	$Y_4 = D$
101	$Y_5 = D$
110	$Y_6 = D$
111	$Y_7 = D$



TRUTH TABLE AND BOOLEAN EQUATIONS

SELECT	BOOLEAN EQUATION	D = 0	D = 1	ACTIVE OUTPUT
S = 000	$Y_0 = D \cdot S_2' \cdot S_1' \cdot S_0'$	00000000	00000001	Y_0
S = 001	$Y_1 = D \cdot S_2' \cdot S_1' \cdot S_0$	00000000	00000010	Y_1
S = 010	$Y_2 = D \cdot S_2' \cdot S_1 \cdot S_0'$	00000000	00000100	Y_2
S = 011	$Y_3 = D \cdot S_2' \cdot S_1 \cdot S_0$	00000000	00001000	Y_3
S = 100	$Y_4 = D \cdot S_2 \cdot S_1' \cdot S_0'$	00000000	00010000	Y_4
S = 101	$Y_5 = D \cdot S_2 \cdot S_1' \cdot S_0$	00000000	00100000	Y_5
S = 110	$Y_6 = D \cdot S_2 \cdot S_1 \cdot S_0'$	00000000	01000000	Y_6
S = 111	$Y_7 = D \cdot S_2 \cdot S_1 \cdot S_0$	00000000	10000000	Y_7

DESIGN FLOW



1:8 DEMUX DESIGN ON VERILOG

Design Source: demux_1to8.v

```
module demux_1to8(input D, input [2:0] S, output reg [7:0] Y);
    always @(*) begin
        Y = 8'b00000000;
        case(S)
            3'b000: Y[0] = D;
            3'b001: Y[1] = D;
            3'b010: Y[2] = D;
            3'b011: Y[3] = D;
            3'b100: Y[4] = D;
            3'b101: Y[5] = D;
            3'b110: Y[6] = D;
            3'b111: Y[7] = D;
        endcase
    end
endmodule
```

TESTBENCH AND VERIFICATION

VERIFICATION METHOD

- Test both $D = 0$ and $D = 1$
- Cycle through all 8 combinations of $S_2S_1S_0$ for both values of D
- Calculate the expected output.
- Compare the expected and actual outputs.
- Display PASS when they match, or FAIL if they don't.

CORE TESTBENCH

```
D = 0;

for (i = 0; i < 8; i = i + 1) begin
    S = i;
    expected = 8'b00000000;

    #10;

    if (Y == expected)
        $display("PASS: D=%b S=%b Y=%b", D, S, Y);
    else
        $display("FAIL: D=%b S=%b Expected=%b Actual=%b",
            D, S, expected, Y);
end

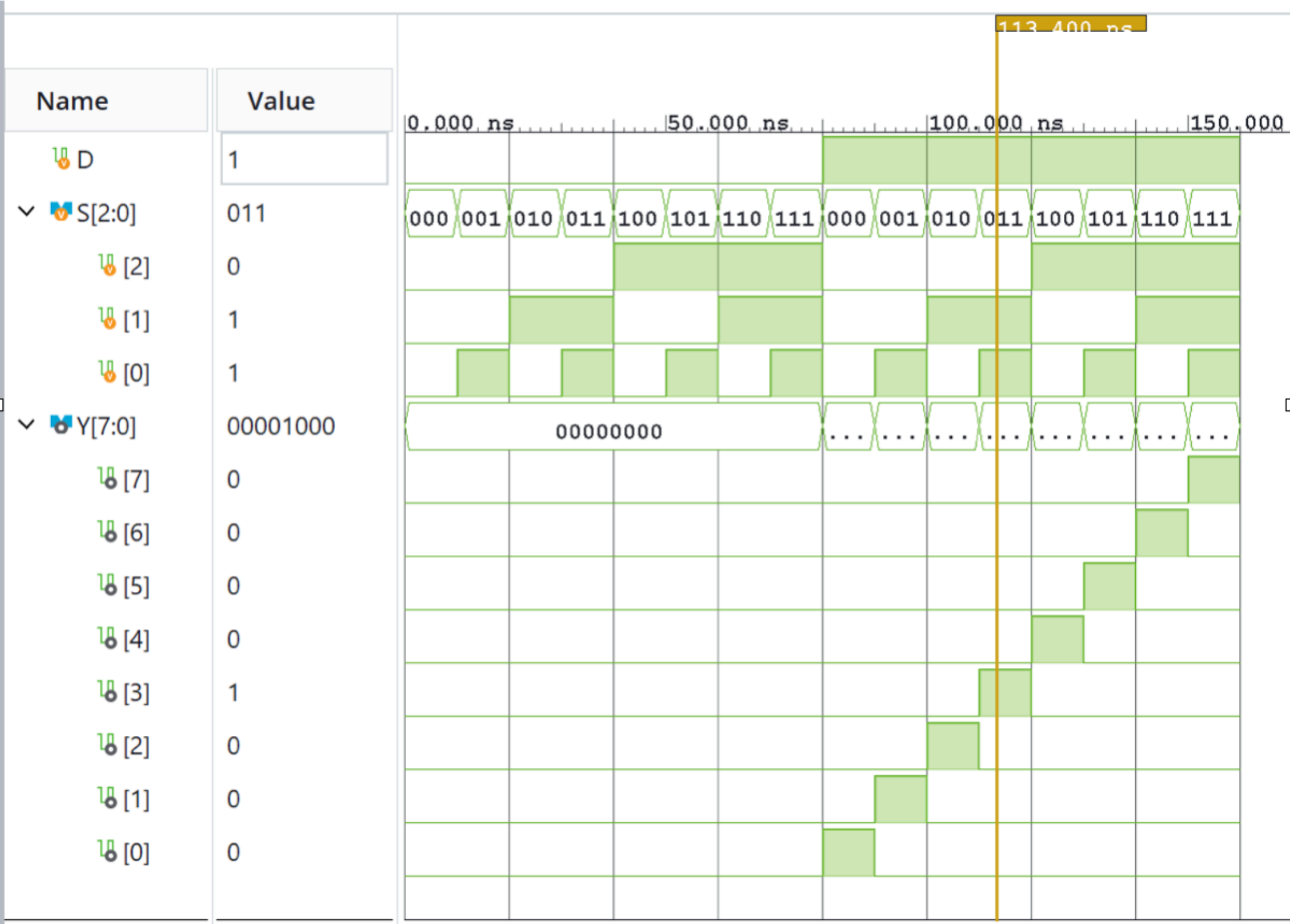
D = 1;

for (i = 0; i < 8; i = i + 1) begin
    S = i;
    expected = 8'b00000000;
    expected[i] = 1'b1;

    #10;

    if (Y == expected)
        $display("PASS: D=%b S=%b Y=%b", D, S, Y);
    else
        $display("FAIL: D=%b S=%b Expected=%b Actual=%b",
            D, S, expected, Y);
end
```

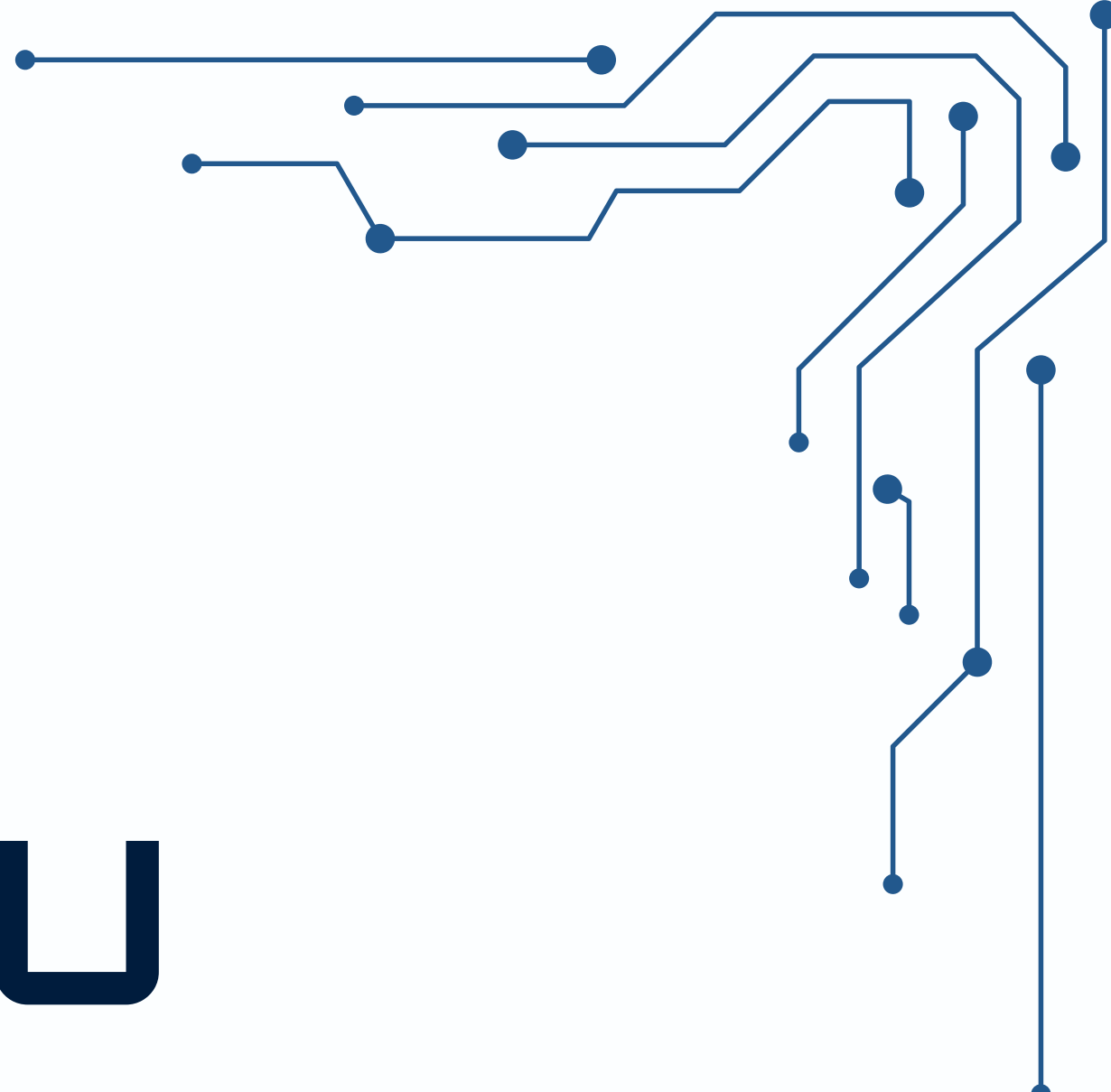
WAVEFORM



Waveform confirms correct routing of D to the selected output.

RESULTS : TCL CONSOLE

```
# run 1000ns
PASS: D=0 S=000 Y=000000000
PASS: D=0 S=001 Y=000000000
PASS: D=0 S=010 Y=000000000
PASS: D=0 S=011 Y=000000000
PASS: D=0 S=100 Y=000000000
PASS: D=0 S=101 Y=000000000
PASS: D=0 S=110 Y=000000000
PASS: D=0 S=111 Y=000000000
PASS: D=1 S=000 Y=000000001
PASS: D=1 S=001 Y=000000010
PASS: D=1 S=010 Y=000000100
PASS: D=1 S=011 Y=000001000
PASS: D=1 S=100 Y=000100000
PASS: D=1 S=101 Y=001000000
PASS: D=1 S=110 Y=010000000
PASS: D=1 S=111 Y=100000000
$finish called at time : 160 ns
```

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THANK YOU